

7.6 Fenchel-Nielsen coordinates

Thm 7.6.3. The map $FN: T_S \rightarrow (\mathbb{R}^*)^r \times \mathbb{R}^r$ is a homeomorphism

Pro 7.6.5. The function FN_T is continuous.

Pf: Consider $\tau_1, \tau_2 \in T_S$, which is close. $\tau_1: \varphi: S \rightarrow X$; $\tau_2: \varphi: S \rightarrow X_\mu$
 $\mu \in M(X)$, $\|\mu\|_\infty$ is small.

Let $X = D/G_1$, μ induces $\tilde{\mu}$ on D . Consider $\bar{\partial} f_{\tilde{\mu}} = \tilde{\mu} \partial f_{\tilde{\mu}}$

$\exists!$ quasi-conformal map s.t. fix $1, -1, i$; when $\|\mu\|_\infty$ small, $f_{\tilde{\mu}}$ near id.

$$\text{Def: } G_2 = f_{\tilde{\mu}} \circ G_1 \circ f_{\tilde{\mu}}^{-1}$$

$$X_\mu \cong D/G_2$$

Now, lift the three geodesics defining $t_r(\tau_1)$:

$\tilde{A}_1$ of A_1 , $\tilde{B}_1$ of B_1 , $\tilde{\varphi}(\gamma_1)$ of $\varphi(\gamma_1)$

and end points are fixed points in G_1

under $f_{\tilde{\mu}}$, they become fixed points in G_2

Since $f_{\tilde{\mu}}$ near id, $\tilde{A}_2 := f_{\tilde{\mu}}(\tilde{A}_1)$ near $\tilde{A}_1$

$\tilde{B}_2 := f_{\tilde{\mu}}(\tilde{B}_1)$ near $\tilde{B}_1$, $\tilde{\varphi}(\gamma_2) = f_{\tilde{\mu}}(\tilde{\varphi}(\gamma_1))$

near $\tilde{\varphi}(\gamma_1)$. Thus, end points change continuously, $\delta_{r_1}, \delta_{r_2}$ as well, so $t_r(\tau_2)$ is closed to $t_r(\tau_1)$ $\square$

Now, lift the three geodesics defining $t_r(\Gamma_1)$:

$\tilde{A}_1$ of A_1 , $\tilde{B}_1$ of B_1 , $\tilde{\varphi}(r_1)$ of $\varphi(r_1)$

and end points are fixed points in G_1

under f_μ , they become fixed points in G_2

Since f_μ near id, $\tilde{A}_2 := f_\mu(\tilde{A}_1)$ near $\tilde{A}_1$

$\tilde{B}_2 := f_\mu(\tilde{B}_1)$ near $\tilde{B}_1$, $\tilde{\varphi}(r_2) = f_\mu(\tilde{\varphi}(r_1))$

near $\tilde{\varphi}(r_1)$. Thus, end points change continuously, $\delta_{r_1}, \delta_{r_2}$ as well, so $t_r(\Gamma_2)$ is closed to $t_r(\Gamma_1)$ $\square$

Proof of 7.6.3: We already shown FN is continuous.

We want to find an inverse map, i.e. $\Psi: (\mathbb{R}_+^*)^\Gamma \times \mathbb{R}^\Gamma \rightarrow T_S$
 s.t. $\Psi \circ FN = \text{id}$, $FN \circ \Psi = \text{id}$.

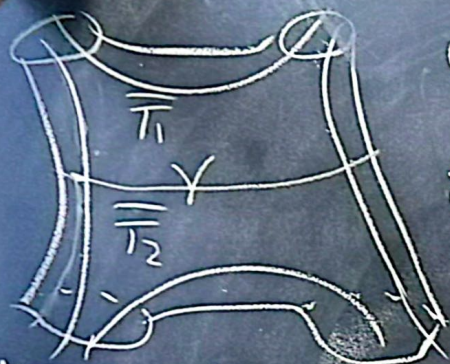
Given $l = (l_r)_{r \in \Gamma}$ by thm 3.5.8, we can construct hyperbolic trousers $\bar{T}_i$ corresponding to the trousers of S defined by Γ .

 $\forall r \in \Gamma$, if r divides ≥ 2 different trousers $\bar{T}_1, \bar{T}_2$

let $r_1 \subset \partial \bar{T}_1, r_2 \subset \partial \bar{T}_2$ boundary circles. length: l_r

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Choose one intersect γ_1, γ_2 , get base points
 x $\leftarrow$ γ_1 , then sew γ_1, γ_2 by $t_\gamma > 0 \rightarrow$
 $< 0 \leftarrow$

If γ divides 1 trouser, its same

$\Rightarrow$ We get $X_{l,t}$ $l = (l_\gamma)_{\gamma \in \Gamma}$
 $t = (t_\gamma)$

$\forall \gamma \in \Gamma$,
 Let coo
 $(\gamma) \leftarrow coo$
 $\gamma \in R/2$
 Curve i
 origin v
 closed g
 And $A_{\eta(l)}$
 $\eta(l) = \frac{1}{2} \ln$
 $\zeta = \zeta + i\eta$

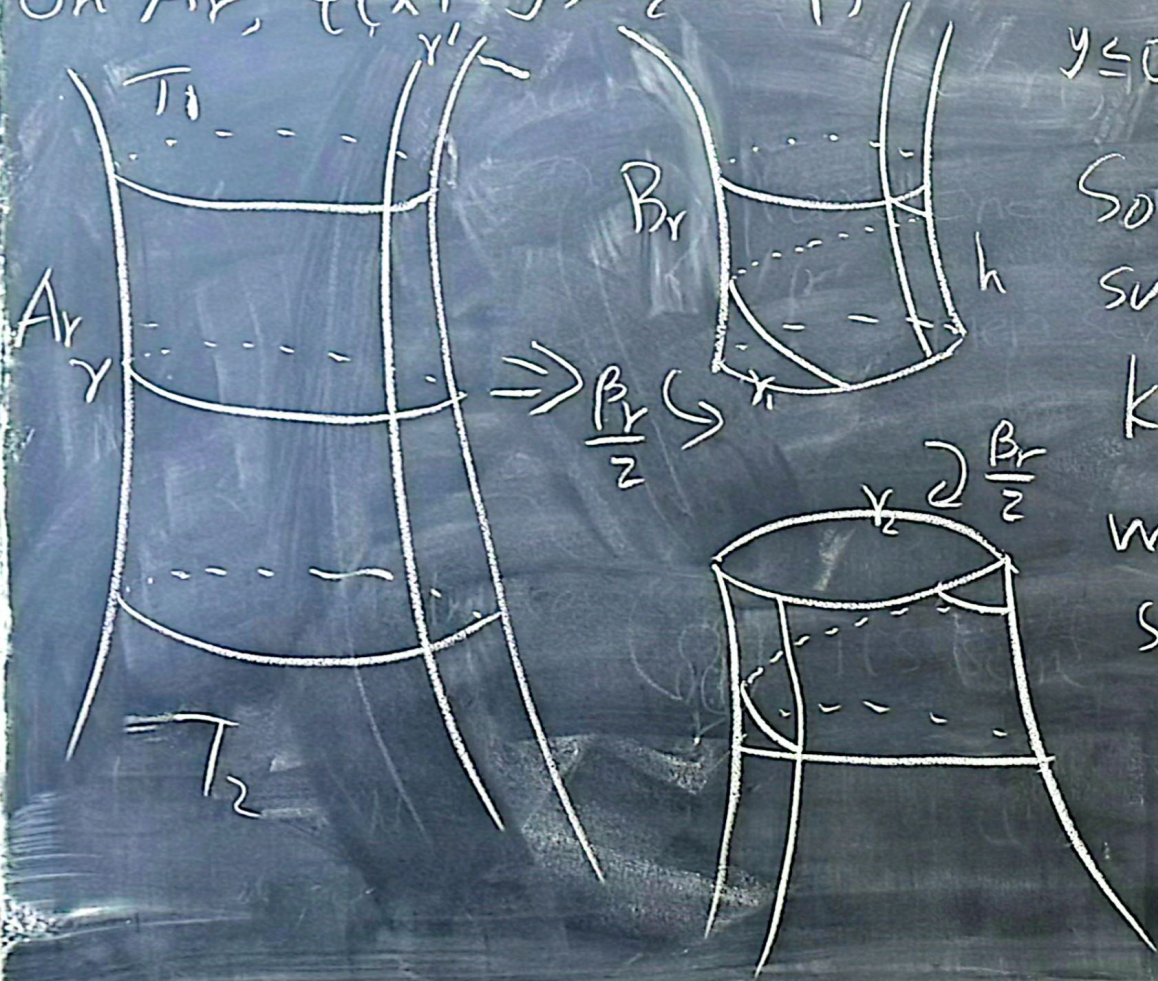
$\forall \gamma \in \Gamma$, choose A_γ as annulus, and disjoint
 Let coordinate $\psi_r: \mathbb{R}/\mathbb{Z} \times [-1, 1] \rightarrow A_\gamma$ s.t.
 $\psi_r(\{0\} \times [-1, 1])$ is ∂A_γ , the other
 curve is $\{1/2\} \times [-1, 1]$ base point p_r is
 origin. $\forall r$, by thm 3.8.3 (collar: γ is a
 closed geodesic, $A_\delta(\gamma) = \{x \in X, d(x, \gamma) < \delta\}$
 And $A_{\eta(l_i)}(\gamma_i)$ are collars, they are disjoint,
 $\eta(l) = \frac{1}{2} \ln \frac{\cosh(\frac{l}{2}) + 1}{\cosh(\frac{l}{2}) - 1}$, we get $B_\gamma \subset X_{L,t}$
 $\zeta = \xi + i\eta$, $\xi \in \mathbb{R}/\mathbb{Z}$, $-h \leq \eta \leq h$, ξ is coordinates

$\forall$ trouser T on S , let $T' = T - \bigcup_{\gamma \in \Gamma} A_\gamma$, trousers shrink
 Remove B_γ , get T'' . Choose a homeomorphism $\varphi_r: T' \rightarrow T''$.
 φ_r maps component of ∂A_γ in T' to components of
 ∂B_γ in T'' , with speed l_γ ; maps Γ' in T' to arcs of
 minimizing geodesics connecting boundary component of
 trousers of $X_{L,t}$.

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On A_r , $\varphi(x+iy) = \xi + i\eta$, when $y \geq 0$, $\xi + i\eta = \left[x + \frac{\beta_r}{2} \cdot \frac{h-y}{y} \right] + i \frac{y}{h}$
 $y \leq 0$, $\xi + i\eta = \left[x - \frac{\beta_r}{2} \cdot \frac{h+y}{y} \right] + i \frac{y}{h}$



So if $t_r \rightarrow t_r + k \cdot l_r$ get same surface, but are different in twist

$$k := \frac{\beta_r - \tilde{\beta}_r}{l_r}$$

With this, we get $\varphi: S \rightarrow X$

So we get Ψ , check $\Psi \circ FN = id$

$$FN \circ \Psi = id$$

